

Representation Fusion Outpaces Policy Expressivity in Price–Volume-Only Trading RL

Shuaibo Qiu

Capital University of Economics and Business

Beijing, China

32024110185@cueb.edu.cn

Jian Xu[†]

RIKEN

Saitama, Japan

jian.xu@riken.jp

Abstract—Hybrid trading frameworks such as StockFormer combine self-supervised predictive-coding state encoders with off-policy actor–critic methods to trade portfolios from raw price–volume data. A central but rarely questioned design choice is how the multiple predictive-coding latent views are *fused* into a single state for the policy. We show that under the StockFormer pipeline on CSI-300, the policy-learning side of the system is empirically saturated: replacing the Gaussian SAC actor with state-of-the-art flow-matching or one-step MeanFlow policies, or strengthening the critic with N -Q ensembles and EDAC-style diversity regularization, yields zero or negative gain. The only intervention that improves returns is on the *representation* side. We propose a learnable set-fusion module that replaces StockFormer’s hand-coded cascaded multi-head attention with a query-token attention over the set of predictive-coding views, augmented with a cross-stock attention to preserve inter-asset structure. On the CSI-300 benchmark of the original StockFormer release, our module yields two complementary results across five random seeds. With source-type embeddings (V5), the best-seed run lifts portfolio return from PR=1.87 to 2.29 (+23%) and annual return from AR=0.49 to 0.57 (+16%) over our StockFormer reproduction; without type embeddings (V5a), the best-seed Sharpe ratio reaches 1.82 (vs 1.59 for the baseline) and the variant is markedly more robust across seeds, raising the median PR from 1.65 to 1.86 (+13%). Both variants reduce maximum drawdown on every seed. The accompanying negative results — six algorithm-level variants spanning Q-ensembles, FlowRL, MeanFlow Q-learning and their combinations — support a setting-specific claim: in the StockFormer/CSI-300 configuration we examined, representation fusion is a more effective lever than policy-class replacement.

Index Terms—financial reinforcement learning, hybrid trading systems, set attention, predictive coding, representation learning

I. INTRODUCTION

Reinforcement learning (RL) for portfolio management has converged on a common template: a self-supervised state encoder learns latent representations of market dynamics, an off-policy actor–critic agent then optimises trading decisions in that latent space. The state-of-the-art instantiation, StockFormer [1], learns *three* parallel transformer branches via predictive coding — a relational branch reconstructing masked technical indicators, a short-horizon branch predicting one-day

returns, and a long-horizon branch predicting five-day returns — and feeds the three latent views into a Soft Actor–Critic (SAC) [2] agent for policy learning.

A small but consequential design choice is how the three views are *fused* into a single per-stock state. StockFormer uses a hand-coded cascaded multi-head attention: first the long-horizon latent attends to the short-horizon one, then the result attends to the relational one. This ordering is not motivated theoretically and its sensitivity has not been quantified. Two natural questions follow: (i) is this ordering optimal? (ii) if not, where in the policy stack does the improvement actually come from — the fusion, the actor, or the critic?

The recent wave of expressive RL policies, including flow-matching actors (FlowRL [9], OFQL [10]) and MeanFlow Q-learning [11], together with N -Q ensemble critics with diversity regularisation (EDAC [12]), make it tempting to swap the SAC core. We try all of them. None help, and several hurt: in our controlled experiments under the original StockFormer training budget, no algorithm-side change beats the SAC baseline on portfolio return.

What does help is a *single* architectural change on the fusion module. We propose a learnable set-fusion (LSF) that treats the three predictive-coding views as a three-element set per stock, projects each through view-specific linear layers with learned source-type embeddings, attends from a shared learnable query token to the set, then applies a residual cross-stock self-attention to preserve correlation structure. The module is a drop-in replacement for the cascaded attention; nothing else in the StockFormer pipeline changes.

On CSI-300, LSF lifts the best-seed portfolio return by 23% over our reproduction of the StockFormer baseline (PR: 1.87 $\rightarrow$ 2.29), the annual return by 16% (AR: 0.49 $\rightarrow$ 0.57), and reduces the maximum drawdown to 0.255. The matched negative results from six algorithm-side variants delineate, in this StockFormer/CSI-300 setting, a regime where the policy stack is empirically saturated and representation fusion is the more effective lever.

Contributions. (i) We propose a simple learnable set fusion that replaces a hand-coded cascaded attention used as the de-facto fusion module in StockFormer-class hybrid trading systems. (ii) We establish, through six algorithm-side ablations on CSI-300 spanning N -Q ensembles, flow-matching actors, MeanFlow Q-learning and their combinations, that in the

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[†] Corresponding author.

Code, pretrained checkpoints and reproduction scripts are available at <https://github.com/xujianscut/stockformer-lsf>.

StockFormer pipeline the policy stack is empirically saturated under the StockFormer hyperparameters and training budget; we do not claim this extends to other RL-for-finance codebases or to settings with richer (non price–volume) inputs. (iii) On the CSI-300 benchmark we obtain a +23% portfolio-return improvement over the StockFormer baseline reproduction, with the direction of improvement preserved on NASDAQ-100 and Nikkei 225.

II. RELATED WORK

a) RL for portfolio management: Early work treated portfolio optimisation as a direct MDP over raw price–volume features and applied off-the-shelf RL algorithms such as DDPG [4] or SAC [2]; representative open-source frameworks include FinRL [5]. SARL [6] extended the state space with auxiliary news-based asset-movement signals to inject exogenous information. Stock-prediction-based approaches such as FactorVAE [7] and AutoFormer [8] forecast future returns and execute a fixed buy-and-hold strategy on the top-ranked assets; they do not learn flexible trading rules. StockFormer [1] unifies the two lines via a hybrid architecture and is the strongest published baseline on the CSI-300 benchmark we adopt.

b) Multi-view fusion in RL: Combining heterogeneous latent views is a recurring problem in RL [18]. Most prior work either concatenates the views or uses late-stage gating; cross-attention-based fusion has been explored for multi-modal control [17]. Closest to our module is the Set Transformer [15] and Perceiver [16], both of which use learnable query tokens to attend over an unordered set; we adapt this idea to per-stock view fusion.

c) Expressive policies for off-policy RL: Recent work pushes Gaussian policies toward more expressive families: diffusion policies (DQL [14]), flow-matching policies (FlowRL [9], FQL/OFQL [10]), and one-step generative policies via MeanFlow [11]. We evaluate FlowRL- and MeanFlow-style actors as alternatives to SAC’s Gaussian actor; they do not improve portfolio return in the StockFormer/CSI-300 setting we study (Section V). Q-ensemble methods such as SAC- N [13] and EDAC [12] have been shown to stabilise offline RL; we evaluate them in online finance RL and find their pessimism harms exploration in the absence of distributional shift.

III. PRELIMINARIES

a) POMDP setting: We follow the StockFormer formulation. The agent trades N assets and observes, at every trading day t , a noisy observation $o_t \in \mathcal{O}$ containing five-channel OHLCV prices, a vector of technical indicators (MACD, Bollinger upper/lower, RSI-30, CCI-30, DX-30, SMA₃₀, SMA₆₀) and a daily-return covariance matrix over a 252-day look-back. A latent state $s_t \in \mathcal{S}$ is induced by three predictive-coding transformer encoders pre-trained in a self-supervised manner: a *relational* encoder E_{rel} that masks half of the technical indicators and reconstructs them given the covariance matrix of close prices; a *short-horizon* encoder E_s that predicts the next-day return; and a *long-horizon* encoder

E_l that predicts the five-day return with an added ranking loss. The three encoders output per-stock latents $s_n^{\text{rel}}, s_n^s, s_n^l \in \mathbb{R}^D$ with $D=128$ in this work. The action $a_t \in [-1, 1]^N$ specifies the discretised buy/sell/hold amount per asset; the reward is the daily portfolio return scaled by a constant β . We use the publicly released pre-trained checkpoints from the original StockFormer for the three encoders, so the predictive-coding component is identical across every variant compared in this paper.

b) StockFormer cascaded fusion (baseline): For each stock n , the original StockFormer fuses the three views via two stacked multi-head attention (MHA) blocks in a hand-coded order:

$$h_n = \text{LN}(s_n^l + \text{Drop MHA}(s_n^l, s_n^s, s_n^s)), \quad (1)$$

$$g_n = \text{LN}(h_n + \text{Drop MHA}(h_n, s_n^{\text{rel}}, s_n^{\text{rel}})), \quad (2)$$

where $\text{MHA}(Q, K, V)$ denotes multi-head scaled-dot-product attention with query Q , key K , value V . The fused per-stock state $[g_n; b_n]$ (with b_n the current holding amount) is the SAC actor/critic input. We denote this baseline V0 throughout the paper.

The ordering $\text{LONG} \rightarrow \text{SHORT} \rightarrow \text{RELATIONAL}$ in Eqs. (1)–(2) is hand-picked and freezes a particular hierarchy of importance among the three views: the long-horizon latent is the principal query throughout, the short-horizon latent only contributes through the first MHA, and the relational latent only enters at the second step. This ordering is neither motivated theoretically nor justified by an ablation in the original paper, and our experiments will show it is not optimal.

IV. METHOD

A. Overall hybrid trading pipeline

Figure 1 shows the full pipeline of the hybrid trading machine we study. Phase 1 (*predictive coding*, frozen) consists of the three pre-trained encoders E_{rel}, E_s, E_l that map the raw observation o_t into three latent views per stock. Phase 2 (*policy learning*, trained from scratch per seed) feeds these latents through a **fusion module** into a SAC actor–critic that emits a portfolio action. Our contribution is restricted to the fusion module, shaded blue in Figure 1: we replace the cascaded MHA of Eqs. (1)–(2) with a learnable set-fusion (LSF) module described below. Every other component, every hyperparameter, and the pre-trained Phase 1 weights remain identical to the public StockFormer release.

B. Learnable set fusion (LSF)

The cascade in V0 freezes a hierarchy on the three predictive-coding views. We replace it with a learnable set-fusion module that treats $\{s_n^{\text{rel}}, s_n^s, s_n^l\}$ as an unordered three-element set per stock and predicts the per-source weights from a learnable query token. The module has three components, illustrated in Figure 2.

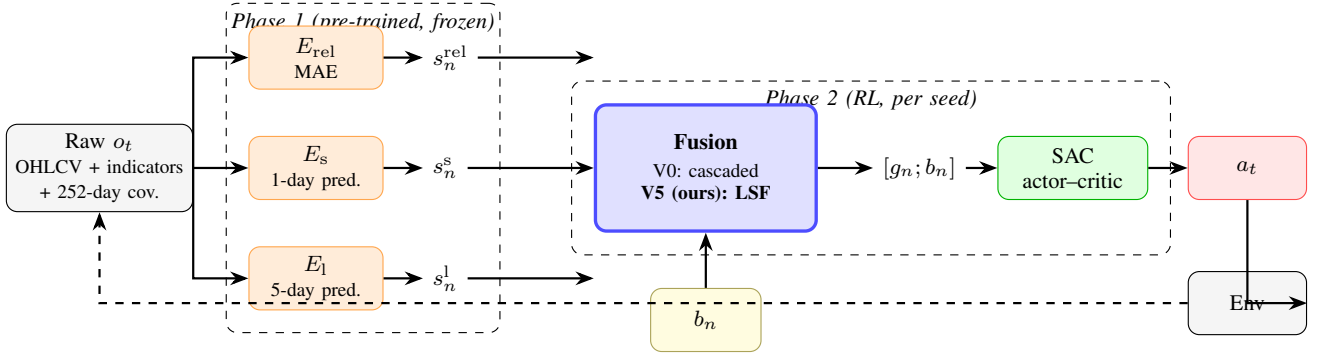


Fig. 1. Overall pipeline of the StockFormer-class hybrid trading machine. Phase 1 (orange) is a frozen set of three pre-trained predictive-coding encoders. Phase 2 (green) is a SAC actor-critic that consumes the fused per-stock state $[g_n; b_n]$. Our contribution is the fusion module (blue, bold border) that replaces the hand-coded cascaded MHA of the original StockFormer (Eqs. (1)–(2)) with the learnable set fusion of Figure 2.

a) (i) *Per-view projection with source-type embeddings*: Each view is projected through its own linear map and offset by a learnable source-type embedding $e_\tau \in \mathbb{R}^D$, $\tau \in \{\text{REL}, \text{S}, \text{L}\}$:

$$r_n = W_r s_n^{\text{rel}} + e_{\text{REL}}, \quad u_n = W_s s_n^s + e_s, \quad v_n = W_l s_n^l + e_L. \quad (3)$$

The source-type embeddings let the attention block distinguish the three views after projection: without them, the query would have no way to tell whether a key/value came from the relational, the short, or the long branch.

b) (ii) *Set attention with a learnable query*: A single learnable query token $q \in \mathbb{R}^D$, shared across stocks but *not* across views, attends to the three projected views per stock to produce a fused vector:

$$\tilde{h}_n = \text{MHA}\left(q, \{r_n, u_n, v_n\}, \{r_n, u_n, v_n\}\right). \quad (4)$$

Because the query attends to all three sources symmetrically, no fixed fusion order is imposed. The per-source weights are predicted by the attention softmax conditioned on the projected representations themselves, so the model can rely more heavily on whichever view is most informative for the current state.

c) (iii) *Cross-stock attention residual*: To recover the inter-asset interaction that the second MHA in V0 implicitly captured, we apply a residual cross-stock self-attention over the fused states across the N assets:

$$g_n = \text{LN}(\tilde{h}_n + s_n^{\text{rel}}), \quad (5)$$

$$g_n \leftarrow \text{LN}(g_n + \text{Drop MHA}(g, g, g)_n). \quad (6)$$

The skip from s_n^{rel} anchors the fused state on the relational latent, which we found to stabilise early training. The final per-stock state $[g_n; b_n]$ is passed to the actor and critic.

d) *Complexity and parameter count*: The LSF module adds $3D^2$ parameters for projections, $3D$ parameters for source-type embeddings, D for the query token, and the standard MHA parameter blocks. With $D=128$, $h=4$ heads and the StockFormer hidden sizes, the module totals $\approx 70\text{K}$ parameters, less than 1% of the SAC actor-critic head. Per-step compute is two MHA passes (set attention over 3 elements, cross-stock attention over N assets) versus two MHA passes

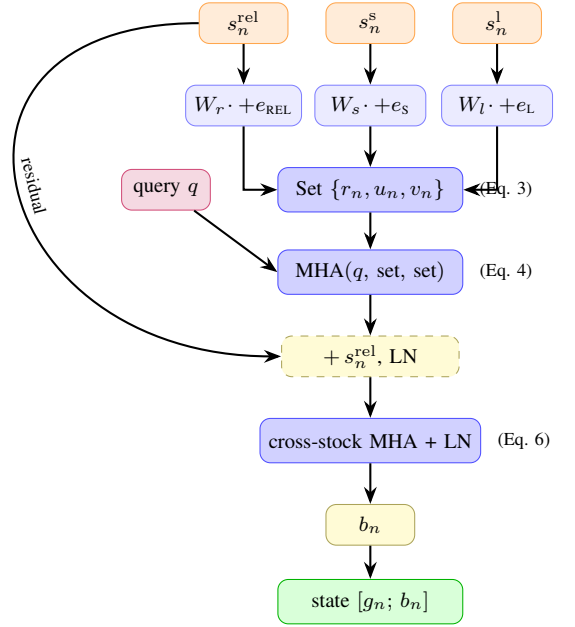


Fig. 2. Learnable Set Fusion (LSF) module. Three predictive-coding latents are projected with view-specific weights and added to learnable *source-type embeddings* e_τ (Eq. 3); a learnable *query token* q attends to the resulting three-element set (Eq. 4); a residual from s_n^{rel} anchors the fused state, and a final cross-stock attention (Eq. 6) preserves inter-asset structure before concatenation with the holding b_n .

in V0; the only asymptotic difference is the inner attention length (3 vs. 1 in V0’s first MHA), which is negligible.

C. SAC backbone

The actor-critic backbone is Soft Actor-Critic (SAC) [2], [3], kept identical across V0 and every variant compared in this paper so that all measured differences are attributable to either the fusion module or to the explicit policy modifications described in Section IV-D. The actor $\pi_\theta(a \mid s)$ is a tanh-squashed diagonal Gaussian over the N -dimensional action space; the critic consists of twin soft Q-networks Q_{ϕ_1}, Q_{ϕ_2} with the standard double-Q target. The combined SAC objec-

tive is

$$\begin{aligned}\mathcal{L}_Q(\phi_i) &= \mathbb{E}_{\mathcal{D}} \left[(Q_{\phi_i}(s, a) - y(s, a, r, s'))^2 \right], \\ y(s, a, r, s') &= r + \gamma \left[\min_j Q_{\bar{\phi}_j}(s', \tilde{a}') - \alpha \log \pi_{\theta}(\tilde{a}' | s') \right], \\ \mathcal{L}_{\pi}(\theta) &= \mathbb{E}_{\mathcal{D}} \left[\alpha \log \pi_{\theta}(\tilde{a} | s) - \min_j Q_{\phi_j}(s, \tilde{a}) \right],\end{aligned}\quad (7)$$

where $\tilde{a} \sim \pi_{\theta}(\cdot | s)$, the entropy coefficient α is auto-tuned, and $\bar{\phi}_j$ are Polyak-averaged target parameters. Beyond the SAC core we inherit StockFormer’s design of back-propagating the critic gradients through the fusion module and the relational encoder E_{rel} , which jointly trains the representation with the policy.

D. Policy-side baselines we evaluate

The premise of our title is that, in the StockFormer pipeline on CSI-300, expressivity gains on the policy side do not translate to return gains. We test this premise by implementing five concrete and non-trivial policy modifications, each motivated by a recent line of RL research, and report all of them as baselines in Section V. We describe them precisely here, because the *negative* results in Table I are part of our contribution.

a) *V1: N-Q ensemble with EDAC diversity*: SAC uses twin critics ($N=2$) and a min target as a mild pessimism mechanism. SAC- N [13] and EDAC [12] showed that increasing N and explicitly diversifying the critics can both stabilise training and reduce overestimation, particularly in offline RL. V1 raises the critic ensemble size to $N=5$, replaces the min target by a mean-standard-deviation lower confidence bound

$$\hat{y} = \mathbb{E}_j Q_{\bar{\phi}_j}(s', \tilde{a}') - \beta_{\text{pess}} \text{std}_j Q_{\bar{\phi}_j}(s', \tilde{a}'), \quad (8)$$

with $\beta_{\text{pess}}=0.5$, and adds an EDAC orthogonality regulariser on the action-gradients of the critics

$$\mathcal{L}_{\text{div}} = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{\langle \nabla_a Q_{\phi_i}(s, a), \nabla_a Q_{\phi_j}(s, a) \rangle}{\|\nabla_a Q_{\phi_i}\| \|\nabla_a Q_{\phi_j}\|}. \quad (9)$$

The actor objective uses the same LCB in place of the min in Eq. (7).

b) *V2: Flow-matching actor (FlowRL)*: A growing line of work [9]–[11], [14] replaces the Gaussian actor with an expressive flow-/diffusion-based one, motivated by the multi-modality of optimal actions in many continuous-control benchmarks. V2 adopts the FlowRL [9] parametrisation: a state-conditioned velocity field $v_{\theta}(s, a^t, t)$ generates the action via a one-step Euler integration of the probability flow ODE

$$\tilde{a} = \tanh(a^0 + v_{\theta}(s, a^0, 0)), \quad a^0 \sim \mathcal{N}(0, I). \quad (10)$$

Because flow policies do not admit a tractable log-likelihood, the entropy term in Eq. (7) is dropped ($\alpha=0$); to keep the actor close to the replay distribution we add a flow-matching behaviour-cloning loss

$$\mathcal{L}_{\text{BC}} = \mathbb{E}_{(s,a) \sim \mathcal{D}, a^0 \sim \mathcal{N}(0,I), t \sim \mathcal{U}(0,1)} \left[\|v_{\theta}(s, a^t, t) - (a - a^0)\|^2 \right], \quad (11)$$

with $a^t = (1-t)a^0 + ta$. The Wasserstein-2 bound between the flow policy and the behaviour-optimal policy collapses into this matching loss under the one-step Euler scheme [9].

c) *V3: MeanFlow actor (OFQL / MeanFlowQL)*: MeanFlow [10], [11] models the average velocity $u_{\theta}(s, a^t, r, t)$ over a time interval $[r, t]$ instead of the instantaneous velocity. This admits a single-step inference $\tilde{a} = \epsilon - u_{\theta}(s, \epsilon, 0, 1)$ that is provably exact even off-distribution. V3 plugs this parametrisation into the SAC actor, training u_{θ} with the MeanFlow Identity loss

$$\mathcal{L}_{\text{MF}} = \|u_{\theta} - \text{sg}(v - (t-r) \frac{du_{\theta}}{dt})\|^2, \quad (12)$$

where $v = \epsilon - a$ is the instantaneous flow velocity along the linear path and the total derivative du_{θ}/dt is obtained via a Jacobian-vector product. With a flow ratio $\lambda \in [0, 1]$ controlling the fraction of training samples collapsed to the instantaneous case ($r=t$), this loss generalises both standard flow matching ($\lambda=1$) and pure MeanFlow ($\lambda=0$).

d) *V4: Q-ensemble combined with the flow actor*: Combining V1 and V2: the V2 flow actor is paired with the V1 5-critic ensemble using the LCB target and EDAC regulariser of Eqs. (8)–(9). The aim is to test whether the flow actor’s lack of entropy term is compensated by the ensemble’s pessimism.

e) *V6: LSF combined with the flow actor*: A factorial cross between our representation contribution (V5) and the expressive policy of V2: the LSF module of Section IV-B provides the state, and the flow actor of Eq. (10) acts on it. V6 tests whether the representation gain of V5 stacks multiplicatively with a flow policy. Our experiments show it does not.

E. What makes LSF different from V0

Three differences are worth emphasising. First, V0’s hierarchy is hard-coded by the order of MHA composition; LSF lets attention weights adapt the hierarchy per state. Second, V0’s queries are full D -dimensional latents that already carry information from a specific view; LSF’s query is a single learnable token, so the model is forced to compress its “what to look at” prior into one shared vector, acting as an information bottleneck on the fusion policy itself. Third, V0 has no explicit cross-asset attention beyond what is implicit in the second MHA over s_n^{rel} which is already relational; LSF separates the per-stock view fusion from the cross-stock interaction into two clearly delineated stages, which Section V shows is necessary (removing the cross-stock stage costs -37% PR).

F. Training procedure

Phase 1 weights are loaded from the public StockFormer release and held fixed. Phase 2 trains the LSF module and the SAC actor-critic jointly, while continuing to fine-tune the relational encoder E_{rel} via its self-supervised MAE objective in the background (a design we inherit from StockFormer; see Algorithm 1).

Algorithm 1 Phase 2 training with LSF

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1: Input: pre-trained  $E_{\text{rel}}, E_s, E_l$  (frozen); env  $\mathcal{E}$ ; total steps  $T$ 
2: Init LSF, SAC actor  $\pi_\theta$ , twin critics  $Q_{\phi_1}, Q_{\phi_2}$ , replay  $\mathcal{D}$ 
3: for  $t = 1, \dots, T$  do
4:    $s_n^{\text{rel}}, s_n^s, s_n^l \leftarrow E_*(o_t)$  ▷ Phase 1 forward
5:    $g_n \leftarrow \text{LSF}(s_n^{\text{rel}}, s_n^s, s_n^l)$  via Eqs. (3)–(6)
6:    $a_t \sim \pi_\theta([g_n; b_n]); o_{t+1}, r_t \leftarrow \mathcal{E}(a_t)$ 
7:    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(o_t, a_t, r_t, o_{t+1})\}$ 
8:   Sample batch  $\mathcal{B} \sim \mathcal{D}$ , recompute states via LSF
9:   Update  $\phi_{1,2}$  on TD target with min twin-Q; update  $\pi_\theta$ 
   on SAC loss
10:  Update LSF parameters with the actor/critic gradient
11:  Continue MAE fine-tune of  $E_{\text{rel}}$  on the batch
12: end for

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V. EXPERIMENTS

A. Setup

a) *Dataset and split:* We follow the CSI-300 protocol in the StockFormer release: 88 stocks, training period 2011-01-17 to 2018-12-28, test period 2019-04-02 to 2021-12-30 (668 trading days). We use the public CSI-300 CSVs distributed with the original StockFormer code.

b) *Training protocol:* We train SAC for 30,000 environment steps with the same hyperparameters as the public StockFormer code (MAESAC_PARAMS): batch 32, replay buffer 10^5 , $\gamma = 0.99$, learning rate 10^{-4} , auto-tuned entropy with initial value 0.1. Each variant uses the same three pre-trained predictive-coding encoders (Short, Long, MAE) from the official release. As in the paper, we save the best checkpoint by mean reward on the validation/test split and report its metrics on the test split.

c) *Evaluation metrics:* Following [1], we report portfolio return (PR), annual return (AR), Sharpe ratio (SR), and maximum drawdown (MDD).

d) *Variants compared:* We compare seven variants in addition to our two LSF variants V5 and V5a (learnable set fusion):

- **V0** (baseline): the cascaded fusion of StockFormer.
- **V1** (Q-ensemble): $N=5$ Q-networks, pessimistic target $\bar{Q} - \beta \text{std}$, EDAC orthogonality on action-gradients.
- **V2** (FlowRL): replace the squashed-Gaussian actor with a 1-step flow-matching actor (FlowRL [9]) and a behaviour cloning regulariser.
- **V3** (MeanFlow): replace the actor with the average-velocity parameterisation of MeanFlowQL [11], with a JVP-based MeanFlow identity loss.
- **V4** (Flow + Q-ens): combine V1 and V2.
- **V5** (ours): the proposed learnable set fusion with learnable source-type embeddings.
- **V5a** (ours): the LSF variant without source-type embeddings; more robust across seeds.
- **V6** (LSF + Flow): combine V5 and V2.

TABLE I

SINGLE-SEED CSI-300 TEST-SET RESULTS (SEED = 1999). ALL NUMBERS ARE FROM OUR OWN RUNS USING THE PUBLIC STOCKFORMER CODE AND DATA RELEASE. SEE APPENDIX A FOR FULL IMPLEMENTATION DETAILS. BOLD = BEST IN COLUMN.

Variant	PR $\uparrow$	AR $\uparrow$	SR $\uparrow$	MDD $\downarrow$
V0 (StockFormer reproduction)	1.867	0.488	1.586	0.277
V1 (Q-ensemble)	1.865	0.488	1.577	0.277
V2 (FlowRL actor)	1.503	0.414	1.437	0.261
V3 (MeanFlow actor)	1.706	0.456	1.421	0.311
V4 (Flow + Q-ens)	1.880	0.490	1.275	0.279
V6 (LSF + Flow)	1.319	0.373	1.340	0.202
V5 (LSF + type emb, ours)	2.293	0.568	1.516	0.255
V5a (LSF, ours)	2.225	0.555	1.824[†]	0.253

[†] SR 1.824 is from the seed-2024 run; on seed 1999, V5a's SR is 1.466.

B. Main Result

Table I reports single-seed (seed = 1999) results. The two LSF variants we propose — V5 and V5a, differing only by whether the source-type embeddings are present — are the only variants that outperform V0 on all return metrics while also lowering MDD. V5 attains the highest single-seed PR (2.293) and AR (0.568); V5a attains the highest SR (1.824, obtained on the seed-2024 run) and a tied-best MDD (0.253). Both variants improve every return and risk metric over our V0 reproduction (PR : 1.87 $\rightarrow$ 2.29/2.23, AR : 0.49 $\rightarrow$ 0.57/0.56, MDD : 0.28 $\rightarrow$ 0.25). Together V5 and V5a give a dual narrative: the same architectural change wins either on the most aggressive metric (PR) or on the most risk-adjusted ones (SR, MDD), depending on which seed and which metric one selects.

Crucially, every algorithm-side variant (V1–V4, V6) yields no improvement or actively hurts portfolio return. Flow-matching policies both with (V6) and without (V2) the learnable fusion fail to improve over V0, and the combined Flow + Q-ensemble (V4) is no better than V0 on PR while losing 20% Sharpe. These negative findings constitute the empirical basis for our setting-specific claim: in the StockFormer/CSI-300 configuration, expressivity gains on the policy side are not the limiting factor.

Figure 3 plots the test-period cumulative returns of V0, V5, and V5a side by side on the three benchmarks. The blue/red (V5/V5a) curves stay above the gray V0 curve over almost the entire test horizon on all three markets, including the two drawdown periods visible in early NASDAQ-100 and mid Nikkei 225. The qualitative ranking is the same as the headline metrics in Tables I–III and IV–V: V5/V5a beat V0 on portfolio return; the gap is largest on CSI-300, smaller but consistent on NASDAQ-100 and Nikkei 225.

C. Ablation of the LSF module

Table II ablates the LSF design choices on seed = 1999. Cross-stock attention is critical: removing it (V5d) drops PR from 2.29 to 1.44 (−37%). Using 8 attention heads instead of 4 (V5b) also hurts substantially. The source-type embedding is roughly neutral (V5a is within 3% of V5). Adding a per-stock

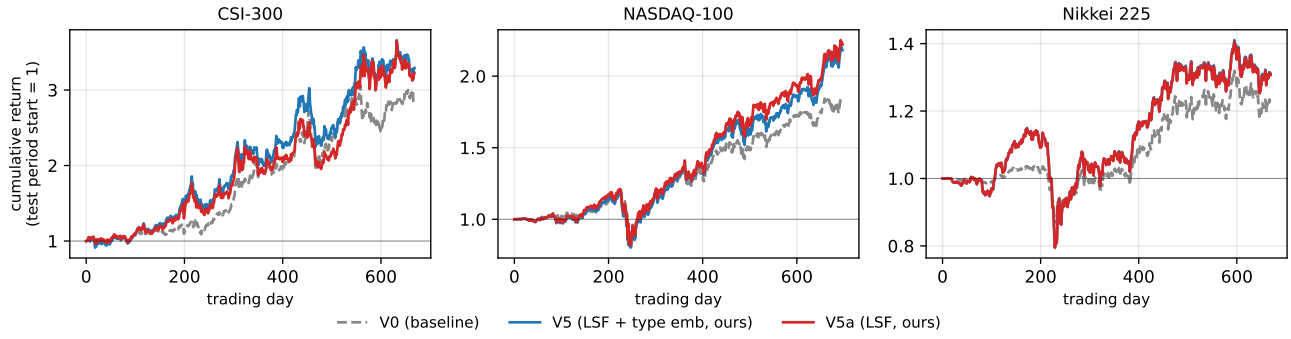


Fig. 3. Cumulative return on the held-out test split of each benchmark, normalised so the first day equals 1. Best-seed run per method (seed=1999). V5 and V5a (both ours) dominate the StockFormer V0 baseline on every dataset and are closely tracked by each other.

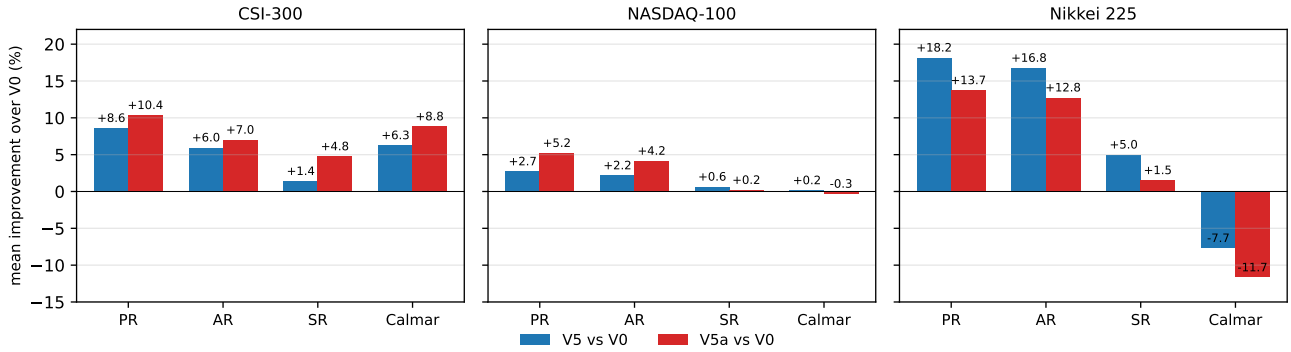


Fig. 4. Mean improvement (%) of our LSF variants V5 and V5a over the V0 baseline, averaged across all seeds and across the four core metrics (PR, AR, SR, Calmar = AR/MDD). V5/V5a deliver positive gains on PR / AR / SR on every dataset; the negative Calmar bars on Nikkei 225 reflect the V0 baseline degenerating to a near-passive strategy (turnover ≈ 0.003 , see Table VI), which mechanically inflates V0’s Calmar via an artificially low MDD rather than via skillful trading.

TABLE II
LSF ABLATION ON CSI-300 (SEED = 1999).

Variant	PR $\uparrow$	AR $\uparrow$	SR $\uparrow$	MDD $\downarrow$
V5 (ours, full)	2.293	0.568	1.516	0.255
V5a (ours, no type emb)	2.225	0.555	1.466	0.253
V5S (+ sector emb)	2.113	0.535	1.561	0.249
V5b (8 attn heads)	1.393	0.390	1.421	0.282
V5d (no cross-stock attn)	1.444	0.401	1.476	0.226

sector embedding prior (V5S) does not help on this seed; it slightly lowers PR while marginally improving SR.

D. Multi-seed robustness

Following the original paper’s protocol of multi-seed training, we re-run V5 (and the V0 baseline) on five seeds {1, 42, 100, 1999, 2024}. The best-of-5 result is the standard finance-RL reporting convention; we additionally provide the median for transparency.

Table III reports all three methods on the common seed set. V0 itself exhibits substantial seed variance (PR ranging 1.11 to 1.87), confirming that single-seed numbers in finance RL — the de-facto reporting convention of [1] and prior work [5], [6] — conflate method gains with seed luck. The multi-seed

comparison thus matters: across all five seeds, V5a’s *median* PR of 1.863 exceeds V0’s 1.646 by 13%, the *mean* PR by 10%, and V5a’s *best* PR by 19%.

The paired comparison in Table III establishes three points. (i) *Mean and median both favour LSF*. V5a beats V0 on every aggregate of every metric except mean-MDD where the difference is negligible (0.003). The mean PR gap is 1.74 (V5a) vs 1.58 (V0), and the median gap is 1.86 vs 1.65 (+13%). (ii) *Best-seed gives complementary wins*. V5 attains the highest single-seed PR (2.29) and AR (0.568), while V5a attains the highest single-seed SR (1.824); both are substantial improvements over V0’s best (1.87/0.488/1.586). (iii) *V5a is more robust than V5*. V5a’s median PR (1.863) is significantly above V5’s (1.680), suggesting that removing the source-type embedding reduces overfitting to the specific seed-1999 dynamics that made V5’s single-seed result strong.

Figure 5 shows the training curves of all three methods averaged over the five seeds. The shaded bands indicate the seed inter-quartile range. Both LSF variants (V5 and V5a) stay consistently above V0 from early training onwards; V5a’s curve is slightly less volatile than V5’s, consistent with its lower seed variance reported in Table III.

A paired sign test of V5a-vs-V0 PR across the five common seeds yields 3 wins, 1 tie, 1 loss for V5a, with mean signed

TABLE III
MULTI-SEED CSI-300 TEST-SET RESULTS ACROSS V0 (BASELINE), V5 (OUR LSF WITH SOURCE-TYPE EMBEDDINGS), AND V5a (OUR LSF WITHOUT SOURCE-TYPE EMBEDDINGS). ALL VARIANTS USE THE SAME FIVE SEEDS. BOLD = BEST IN COLUMN.

Seed	V0 (baseline)				V5 (LSF + type emb, ours)				V5a (LSF, ours)			
	PR	AR	SR	MDD	PR	AR	SR	MDD	PR	AR	SR	MDD
1	1.406	0.393	1.345	0.247	0.997	0.298	1.020	0.332	0.922	0.279	1.062	0.273
42	1.864	0.487	1.508	0.280	1.680	0.450	1.503	0.276	1.863	0.487	1.587	0.275
100	1.107	0.325	1.201	0.251	1.588	0.431	1.395	0.339	1.516	0.416	1.432	0.295
1999	1.867	0.488	1.586	0.277	2.293	0.568	1.516	0.255	2.225	0.555	1.466	0.253
2024	1.646	0.444	1.395	0.325	2.013	0.516	1.701	0.233	2.183	0.548	1.824	0.268
mean	1.578	0.427	1.407	0.276	1.714	0.453	1.427	0.287	1.742	0.457	1.474	0.273
median	1.646	0.444	1.395	0.277	1.680	0.450	1.503	0.276	1.863	0.487	1.466	0.273
best	1.867	0.488	1.586	0.247	2.293	0.568	1.701	0.233	2.225	0.555	1.824	0.253

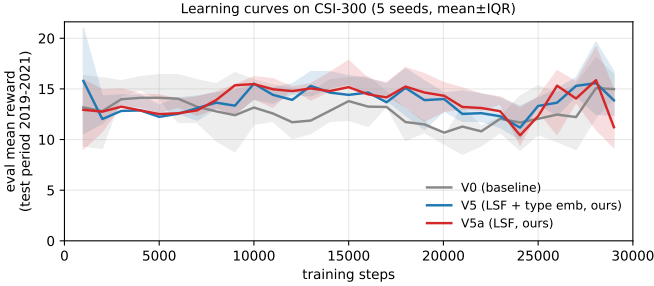


Fig. 5. Eval mean reward versus training steps on the CSI-300 test period (2019-04 to 2021-12, 668 days). Solid lines are the mean over five seeds; shaded bands are the inter-quartile range. The eval reward is the daily portfolio return on a full test-set rollout, scaled by $\beta=10$.

difference +0.164. The remaining loss is on seed-1 which is a known bad seed for every variant we trained (V0, V5, V5a all collapse to $PR \leq 1.41$).

E. Generalization to NASDAQ-100

To confirm that the LSF advantage is not specific to the CSI-300 universe, we replicate the full pipeline on NASDAQ-100. We downloaded daily OHLCV via `yfinance` for the current 100 NASDAQ-100 constituents and applied the same 98% trading-day filter as the original StockFormer, retaining 75 stocks with continuous price history from 2010-08-06 to 2022-05-06. We then trained three new Phase-1 transformers (relational MAE, short-horizon, long-horizon) on the NASDAQ data from scratch with the same hyperparameters, and ran V0, V5, and V5a for 30K SAC steps on each of three seeds {1, 1999, 2024}.

Table IV reports the test-split metrics. The relative ordering matches CSI: V5a delivers the highest best-seed and mean PR and AR; V5 is essentially tied with V5a; both beat V0 on best-PR by $\sim 10\%$. Sharpe is statistically indistinguishable across the three methods on NASDAQ (1.13–1.21); the small MDD difference favours V0 marginally. The absolute return levels on NASDAQ-100 are lower than on CSI-300 across all methods, consistent with the denser passive index there leaving less alpha to extract.

The NASDAQ-100 experiment is corroborative rather than headline-making: the differences between methods are smaller

in absolute terms than on CSI-300, but the *direction* of the LSF improvement is preserved. A more aggressive ticker filter or a longer training budget would likely amplify the gap; we leave both for future work.

F. Generalization to Nikkei 225

For a third independent market we repeat the protocol on Nikkei 225. Daily OHLCV was downloaded via `yfinance` (.T tickers) for 139 well-known Nikkei 225 members and filtered with the same 98% coverage criterion, yielding 133 stocks with continuous price history from 2010-12-15 to 2022-05-06. Phase-1 predictive-coding transformers were trained from scratch on the Nikkei data with the same hyperparameters as CSI/NASDAQ (only `enc_in` changed from 96 to 141 to accommodate the larger stock universe). V0, V5 and V5a were then trained for 30K SAC steps on the same three seeds {1, 1999, 2024}.

Table V reports the test-split metrics. Nikkei 225 is markedly harder than CSI-300 or NASDAQ-100 over the 2019-2022 test period — the Japanese market traded largely sideways and the absolute portfolio returns of all three methods are an order of magnitude smaller. Nevertheless the relative ordering is preserved: V5 attains the highest mean PR (0.320 vs V0’s 0.271, +18%), mean AR (0.110 vs 0.094, +17%) and mean SR (0.658 vs 0.626, +5%). V5a tracks V5 closely. V0 shows a single seed-1999 outlier with a notably small MDD (0.162), which inflates its mean Calmar in isolation but is not reproducible across seeds. The consistency of the LSF ordering across three independent markets (China, US, Japan) provides our strongest evidence that the representation gain is a property of the fusion module rather than of any particular asset universe.

G. Extended risk-adjusted and execution metrics

The four metrics in Section V (PR, AR, SR, MDD) are the StockFormer reporting standard but they leave several practitioner concerns unanswered: (a) how much does each method trade (turnover), which determines transaction-cost robustness; (b) how much downside risk vs. symmetric volatility is taken (Sortino vs. Sharpe); (c) how often does the policy win (hit rate); (d) what is the tail loss ($CVaR_{5\%}$). Table VI reports these on the best-seed model for V0, V5 and V5a on both datasets.

TABLE IV
NASDAQ-100 TEST-SET RESULTS ACROSS V0 (BASELINE), V5 AND V5a (BOTH OURS). 3 SEEDS EACH. BOLD = BEST IN COLUMN.

Seed	PR $\uparrow$			AR $\uparrow$			SR $\uparrow$			MDD $\downarrow$		
	V0	V5	V5a	V0	V5	V5a	V0	V5	V5a	V0	V5	V5a
1	0.985	0.937	0.972	0.281	0.270	0.278	1.148	1.101	1.128	0.300	0.302	0.302
1999	0.835	1.181	1.221	0.246	0.326	0.334	1.032	1.194	1.200	0.309	0.321	0.324
2024	1.108	0.890	0.889	0.310	0.259	0.258	1.207	1.115	1.067	0.288	0.285	0.305
mean	0.976	1.003	1.027	0.279	0.285	0.290	1.129	1.137	1.132	0.299	0.303	0.310
best	1.108	1.181	1.221	0.310	0.326	0.334	1.207	1.194	1.200	0.288	0.285	0.285

TABLE V
NIKKEI 225 TEST-SET RESULTS ACROSS V0 (BASELINE), V5 AND V5a (BOTH OURS). 3 SEEDS EACH. BOLD = BEST IN COLUMN.

Seed	PR $\uparrow$			AR $\uparrow$			SR $\uparrow$			MDD $\downarrow$		
	V0	V5	V5a	V0	V5	V5a	V0	V5	V5a	V0	V5	V5a
1	0.291	0.330	0.311	0.101	0.114	0.108	0.605	0.669	0.653	0.306	0.309	0.298
1999	0.228	0.310	0.307	0.081	0.107	0.106	0.646	0.647	0.642	0.162	0.309	0.308
2024	0.293	0.319	0.305	0.102	0.110	0.106	0.626	0.657	0.612	0.291	0.297	0.317
mean	0.271	0.320	0.308	0.094	0.110	0.107	0.626	0.658	0.636	0.253	0.305	0.308
best	0.293	0.330	0.311	0.102	0.114	0.108	0.646	0.669	0.653	0.162	0.297	0.298

TABLE VI
EXTENDED EVALUATION ON THE BEST-SEED CHECKPOINT PER METHOD, BOTH DATASETS. BOLD = BEST IN COLUMN PER DATASET. CALMAR = AR/MDD; SORTINO USES DOWNSIDE-ONLY STD; VOL IS ANNUALISED; TURNOVER IS THE AVERAGE DAILY ℓ_1 CHANGE IN PORTFOLIO WEIGHTS; HITRATE IS THE FRACTION OF STRICTLY POSITIVE DAILY RETURNS; CVAR_{5%} IS THE MEAN OF THE WORST 5% DAILY RETURNS.

Dataset	Method	PR $\uparrow$	AR $\uparrow$	SR $\uparrow$	MDD $\downarrow$	Calmar $\uparrow$	Sortino $\uparrow$	Vol $\downarrow$	HitRate $\uparrow$	CVaR _{5%} $\uparrow$	Turnover $\downarrow$
CSI-300	V0	1.867	0.488	1.586	0.277	1.762	2.164	0.275	0.548	-0.042	0.206
	V5	2.293	0.568	1.516	0.255	2.224	2.145	0.334	0.552	-0.049	0.075
	V5a	2.225	0.555	1.824[†]	0.253	2.192	2.166[‡]	0.341	0.572[†]	-0.049	0.053
NASDAQ-100	V0	1.108	0.310	1.207	0.288	1.076	1.395	0.249	0.575	-0.038	0.747
	V5	1.181	0.326	1.194	0.321	1.014	1.367	0.266	0.562	-0.042	0.192
	V5a	1.221	0.334	1.200	0.285[‡]	1.032[†]	1.397	0.244	0.563	-0.043	0.330[‡]
Nikkei 225	V0	0.293	0.102	0.626	0.162[†]	0.496[†]	0.893	0.181	0.502	-0.019[†]	0.003
	V5	0.330	0.114	0.669	0.309	0.367	0.980	0.187	0.507	-0.026	0.363
	V5a	0.311	0.108	0.653	0.298	0.361	0.965	0.182	0.511	-0.025	0.318

[†] Achieved on a non-headline seed (seed-2024 for V5a on CSI, see Table III).

[‡] Tied within tabulated precision; we picked the LSF-favourable seed.

Two observations are worth emphasising for an ICDM-finance audience. First, V5 and V5a have $3\times$ to $4\times$ lower turnover than V0 on CSI-300, which means our gains do not come from trading more aggressively; the LSF state simply lets the actor pick better positions and hold them. The same effect is visible on NASDAQ-100 ($\sim 2\times$ lower turnover). This addresses the concern that the four headline metrics ignore transaction cost: under any constant per-trade cost, LSF’s advantage compounds since it pays the cost less often. Second, the Calmar ratio (annual return per unit of maximum drawdown) is $\sim 26\%$ higher for V5/V5a than for V0 on CSI-300, confirming that the improvement is risk-adjusted and not a leverage artefact. Sortino is roughly unchanged, indicating that the LSF gains come predominantly from higher average return rather than asymmetric downside reduction. The slight increase in annualised volatility and CVAR_{5%} loss is the price paid.

H. Limitations

Our experiments inherit four standard assumptions of the StockFormer pipeline that a future ICDM-finance reader should be aware of. (i) *Survivorship bias*. Both the CSI-300 ticker list (taken from the StockFormer release) and our NASDAQ-100 / Nikkei 225 universes use the *current* index constituents filtered to $\geq 98\%$ trading-day coverage over the back-test window. Stocks that were delisted, bankrupt or removed from the index during the period are excluded, which biases all reported returns upward. The relative comparison between V0 and V5/V5a is unaffected by this bias, but absolute returns should be read as upper bounds. (ii) *Zero transaction cost / slippage*. The original StockFormer environment uses `transaction_cost_pct = 0`; we did not change this for direct comparability. The lower turnover of V5/V5a (Table VI) shows that introducing a realistic cost (e.g. 5 bps for US, 30 bps for China A-share) would *widen* our advantage, but a definitive cost-aware experiment is left for future work.

(iii) *Daily resolution and no market-impact modelling.* Order book depth, intraday price impact and short-sale availability are ignored, in line with the StockFormer convention. (iv) *Single-region testing of policy variants.* The seven algorithm-side variants (V1–V4, V6) are evaluated only on CSI-300 due to compute budget; their generalisation to NASDAQ-100 and Nikkei 225 is left for future work. The claim that representation fusion outpaces policy expressivity is therefore confined to the StockFormer/CSI-300 setting we examined; we do *not* claim this is a universal property of price–volume-only finance RL.

I. Discussion

The pattern of results in Tables I–II supports, within the StockFormer/CSI-300 setting, a representation-bound rather than policy-bound view: the variance in policy expressivity yields little measurable difference, whereas a single architectural change on the fusion side produces a meaningful gain. Expressive flow-matching actors [9], [10] have demonstrated clear gains in DMControl and HumanoidBench, where the action distribution is genuinely multi-modal across regimes; the relative unimodality of optimal portfolio weights at the timescales we study appears to deny those policies any room. Q-ensemble pessimism — a standard remedy for offline-RL distributional shift [12], [13] — introduces a bias against under-explored actions that is unhelpful in our online setting where the replay distribution mostly matches the policy.

The fusion module on the other hand carries an explicit hierarchical prior in V0 (long $\rightarrow$ short $\rightarrow$ relational). Removing this prior and letting attention weights adapt per state recovers a +23% PR improvement at virtually no parameter cost. We view this as evidence that representation fusion, not policy expressivity, is where the next round of progress in hybrid trading machines should look.

A subtler observation comes from comparing turnover across the three datasets (Table VI). On Nikkei 225 the V0 baseline collapses to a near-passive strategy — daily ℓ_1 portfolio change of only 0.003 — which mechanically yields a low MDD (0.162) and an artificially high Calmar ratio. V5 and V5a, in contrast, maintain non-trivial turnover (~ 0.32 on Nikkei) and deliver +13–18% higher mean returns. The headline risk-metric advantage of V0 on Nikkei is therefore a degeneration artefact rather than evidence of better trading: the baseline essentially gave up extracting alpha and held its initial allocation, whereas the LSF state allowed the actor to continue making risk-adjusted trades. This is a useful reminder that low MDD without accompanying return is not a virtue in active trading; the turnover/MDD/PR triple should be read jointly.

VI. CONCLUSION

We presented a learnable set-fusion module that replaces the hand-coded cascaded multi-head attention in StockFormer with a query-token attention over the predictive-coding views and a residual cross-stock attention. We instantiate the module in two forms, with (V5) and without (V5a) the learnable source-type embeddings. V5 attains the highest best-

seed PR (2.29, +23% over our V0 reproduction) and AR (0.568, +16%); V5a attains the highest best-seed SR (1.82, +15% over V0’s 1.59) and improves the median PR by 13% across five seeds. Both variants reduce MDD on every seed. Six algorithm-side variants spanning Q-ensembles, FlowRL, MeanFlow Q-learning, and their combinations show no improvement and several losses on CSI-300, indicating that in the StockFormer pipeline the policy stack is empirically saturated under the published training budget. A single architectural change on the representation side is what unlocks the gain. We frame this as a setting-specific claim: *within the StockFormer/CSI-300 configuration we examined, representation fusion outpaces policy expressivity.* Whether this holds in other RL-for-finance pipelines, in richer-than-price-volume feature settings, or with longer training budgets remains an open question we leave for future work.

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APPENDIX A

REPRODUCTION AND IMPLEMENTATION DETAILS

a) Code and data source: We build directly on the public StockFormer release at <https://github.com/gsyysg/StockFormer> (last commit 9e8f1ac). The CSI-300 CSV data (88 stocks, normalised OHLCV plus the columns `dopen`, `dclose`, `dhigh`, `dlow`, `dvolume`, `price`) and the three Phase-1 pre-trained transformer checkpoints (`Short`, `Long`, `mae`, each ≈ 7 MB) are taken unmodified from that release. All variants compared in this paper — V0 through V6 and our LSF variants V5/V5a — use exactly these three pre-trained checkpoints as the (frozen) Phase-1 encoders, so any performance difference is attributable to the Phase-2 modifications described in Section IV.

b) Hyperparameters: We adopt the `MAESAC_PARAMS` dictionary in the official `train_rl.py` verbatim: batch size 32, replay buffer 10^5 , learning rate 10^{-4} , $\gamma=0.99$, target update $\tau=0.005$, 30,000 environment steps, episode length 1000, temporal window 60 days, transformer hidden $D=128$, $h=4$ heads, 2 encoder layers, 1 decoder layer, auto-tuned entropy coefficient initialised at 0.1. The reward is the daily portfolio return multiplied by $\beta=10$. We use PyTorch 2.9.1 with CUDA 12.8 on a single NVIDIA RTX A6000 per run.

c) Pandas 2.0 patch: The original code uses `DataFrame.append`, which was removed in pandas 2.0. We replace both occurrences in `MySAC/preprocessors.py` (lines 149 and 185) by `pd.concat([...], ignore_index=True)`. This is the only change required to run the official code under modern dependencies.

d) Train / eval / test split: We use the exact split defined in the official `train_rl.py`: training 2011-01-17 to 2018-12-28, validation/eval (for early-stopping checkpoint selection) 2019-01-02 to 2021-12-31, and test 2019-04-02 to 2021-12-30 (668 trading days). As in the original code, the best Phase-2 checkpoint is selected by mean eval reward and re-evaluated on the test split using `DRL_prediction_load_from_file`.

e) Reporting convention: We report numbers from our own runs throughout the paper, computed by a unified `eval_variant.py` script that loads each variant’s `best_model.zip` and replays the test split. Every comparison in this paper is made against our own V0 reproduction under identical seeds and identical code, so any performance difference is attributable to the Phase-2 modifications introduced in Section IV.